Notes on algebraic information theory

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Abstract

This article consists of some notes taken by the author while studying algebraic information theory.

References: [1], [2]

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1 The category of information structures

1.1 Terminology and examples

Here we call random variables (r.v) on a finite set Ω **congruent** when they define the same partition (remind that a partition of Ω is a family of disjoint non-empty subsets covering Ω and that the partition associated to a r.v X is the family of subsets $\Omega_x \subset \Omega$ defined by the equations $X(\omega) = x$).

Let Ω be a finite set, the set $\Pi(\Omega)$ of all partitions of Ω constitutes a category with one arrow $Y \to Z$ from Y to Z when Y is more fine than Z, we also say in this case that Y divides Z.

In $\Pi(\Omega)$ we have an **initial element**, which is the partition by points, denoted ω and a **final(terminal) element**, which is Ω itself and is denoted by 1.

The **joint partition** YZ or (Y, Z), of two partitions Y, Z of Ω is the less fine partition that divides Y and Z, i.e., their gcd.

For any X we get $XX = X$, $\omega X = \omega$ and $1.X = X$, which implies each partition is idempotent.

From above properties of the partitions, we can strictly give the definition of **information structure**.

Definition 1.1 (Partially ordered set)**.** *A **reflexive, weak** or **non-strict partial order** commonly referred to simply as a **partial order**, is a homogeneous relation $\leq$ on a set P that is reflexive, antisymmetric, and transitive. That is, for all $a, b, c \in P$, it must satisfy:*

1. ***Reflexivity**: $a \leq a$, i.e. every element is related to itself.*
2. ***Antisymmetry**: if $a \leq b$ and $b \leq a$ then $a = b$, i.e. no two distinct elements precede each other.*
3. ***Transitivity**: if $a \leq b$ and $b \leq c$ then $a \leq c$.*

Definition 1.2 (meet-semilattice)**.** *A set S partially ordered by the binary relation $\leq$ is a **meet-semilattice** if for all elements x and y of S, the greatest lower bound of the set $\{x, y\}$ exists.*

*The greatest lower bound of the set $\{x, y\}$ is called the **meet** of x and y, denoted $x \wedge y$.*

Replacing "greatest lower bound" with "least upper bound" results in the dual concept of a **join-semilattice**. The least upper bound of $\{x, y\}$ is called the **join** of x and y, denoted $x \vee y$.

By definition an **information structure** $\mathcal{S}$ on Ω is a subset of $\Pi(\Omega)$, such that for any element X of $\mathcal{S}$, and any pair of elements Y, Z in $\mathcal{S}$ that X refines, the joint partition YZ also belongs to $\mathcal{S}$. In addition we will always assume that the final partition 1 belongs to $\mathcal{S}$. In terms of observations, it means that at least something is a certitude.

Definition 1.3 (conditional meet semilattice is a poset)**.** *A conditional meet semilattice is a poset that satisfies the following property:*

for any $X, Y, Z \in Ob\,\mathbf{S}$, if $Z \to X$ and $Z \to Y$, then the categorical product $X \wedge Y$ exists.

It is unital whenever it has a terminal object, denoted $\top$.

Definition 1.4 (information structure)**.** *An information structure is a pair $(\mathbf{S}, \mathcal{M})$, where $\mathbf{S}$ is a **unital conditional meet semilattice** and $\mathcal{M} : \mathbf{S} \to \mathrm{Meas}_{\mathrm{surj}}$ is a functor (say $\mathcal{M}(X) = \mathcal{M}_X = (\mathscr{E}_X, \mathscr{B}_X)$, for each $X \in \mathrm{Ob}\,\mathbf{S}$) that satisfies:*

1. *$\mathscr{E}_\top \cong \{*\}$, with the trivial σ-algebra;*
2. *for every $X \in \mathrm{Ob}\,\mathbf{S}$ and any $x \in \mathscr{E}_X$, the σ-algebra $\mathscr{B}_X$ contains the singleton $\{x\}$;*
3. *for every diagram $X \xleftarrow{\pi} X \wedge Y \xrightarrow{\sigma} Y$ the measurable map*

$$\mathcal{M}_{X \wedge Y} \hookrightarrow \mathcal{M}_X \times \mathcal{M}_Y,\ z \mapsto (x(z), y(z)) := (\mathcal{M}_\pi(z), \mathcal{M}_\sigma(z))$$

is an injection.

Example 1.1. *Start with a set $\Sigma = \{S_i; 1 \leq i \leq n\}$ of partitions of Ω. For any subset $I = \{i_1, ..., i_k\}$ of $[n] = \{1, ..., n\}$, the joint $(S_{i_1}, ..., S_{i_k})$, also denoted S_I, divides each S_{i_j}. The set $W = W(\Sigma)$ of all the S_I, when I describes the subsets of $[n]$ is an information struture. It is even a commutative monoid, because any product of elements of W belongs to W, and the partition associated to Ω itself gives the identity element of W. The product $S_{[n]}$ of all the S_i is maximal; it divides all the other elements. As $\Pi(\Omega)$ the monoid $W(\Sigma)$ is idempotent, i.e., for any X we have $XX = X$.*

An information structure $(\mathbf{S}, \mathcal{M})$ is said to be **bounded** if the poset $\mathbf{S}$ has finite height. It is **finite** if all the sets $\mathscr{E}_X$ are finite, in which case $\mathscr{E}_X$ corresponds to the atoms of $\mathscr{B}_X$ and the algebra can be omitted from the description. We denote a finite structure by $(\mathbf{S}, \mathscr{E})$, where $\mathscr{E}$ is a covariant functor from $\mathbf{S}$ to $\mathbf{Sets}$. The cohomological computations concern finite structures, but the general constructions do not require this hypothesis. In fact, they only depend on the combinatorial object $\mathbf{S}$.

Example 1.2. *Given a set Ω, let $\mathbf{Obs}_{\text{fin}}(\Omega)$ be the category **finite observables**; the objects of this category are finite partitions of Ω, and there is an arrow $X \to Y$ whenever X refines Y. In this case, X discriminates better between the **configurations** $w \in \Omega$. The category $\mathbf{Obs}_{\text{fin}}(\Omega)$ has a terminal object: the trivial partition $\top := \{\Omega\}$. When Ω is finite, it also has an initial object: the partition by points, that we denote by $\perp$. The categorical product $X \times Y$ of two partitions X and Y is the coarsest partition that refines both. This product is commutative, associative, idempotent and unitary ($\top \times X = X$).*

A **classical information structure** in the sense of [2] is a full subcategory $\mathbf{S}$ of $\mathbf{Obs}_{\text{fin}}(\Omega)$ such that

- $\top \in \operatorname{Ob} \mathbf{S}$;
- for any X, Y, Z in $\operatorname{Ob} \mathbf{S}$, if $X \to Y$ and $X \to Z$, then $Y \times Z$ belongs to $\mathbf{S}$.

We call $\mathbf{S}$ a **concrete structure**. If $\square : \mathbf{Obs}_{\text{fin}}(\Omega) \to \mathbf{Sets}$ denotes the "forgetful" functor that maps the partition $X = \{A_1, ..., A_n\}$ to the set $\{A_1, ..., A_n\}$ and each arrow $X \to Y$ in $\mathbf{Obs}_{\text{fin}}(\Omega)$ to the unique surjective map $\square\pi : \mathscr{E}_X \to \mathscr{E}_Y$ such that $B = \bigcup_{A \in \mathscr{E}_{\pi^{-1}(B)}} A$ for any $B \in \mathscr{E}_Y$, the pair $(\mathbf{S}, \square)$ is a finite information structure according to the definition of information structure.

Concrete structures turn out to be too restrictive. For instance, Baudot and Bennequin associate to any finite indexed collection $\Sigma = (S_1, ..., S_n)$ of partitions of Ω a "simplicial structure" $\mathbf{S}(\mathbf{K})$: a subcategory of $\mathbf{Obs}_{\text{fin}}(\Omega)$ that contains $\prod_{i \in A} S_i$ for any object A of a simplicial subcomplex $\mathbf{K}$ of the abstract simplex $\Delta(\{1, ..., n\})$; by convention, the empty product gives the trivial partition. Such construction does not necessarily give an information structure (in their sense). For example: if $n = 3$, $\Omega = \{0, 1\}^2$, S_i is the partition induced by the projection on the i-th component $(i = 1, 2)$, $S_3 = \{\{(0, 0)\}, \{(0, 1)\}\}$, and the maximal cells of $\mathbf{K}$ are $\{1, 2\}$ and $\{3\}$, then

$S_1 \times S_2$ is the atomic partition, that refines all the others, while some products (like $S_1 \times S_3$) are not in $\mathbf{S}(\mathbf{K})$. In our framework, the category $\mathbf{S}(\mathbf{K})$ appears as a classical representation of the (generalized) information structure $(\mathbf{K}, \mathscr{E})$, where $\mathscr{E} : \mathbf{K} \to \mathbf{Sets}$ is given by $\mathscr{E}_{\{1\}} = \mathscr{E}_{\{2\}} = \{0, 1\}$, $\mathscr{E}_{\{1,2\}} = \mathscr{E}_{\{1\}} \times \mathscr{E}_{\{2\}}$, the maps induced by the arrows in $\mathbf{K}$ being canonical projections.

References

[1] Juan Pablo Vigneaux. "Information structures and their cohomology". In: *arXiv preprint arXiv:1709.07807* (2017).

[2] Pierre Baudot and Daniel Bennequin. "The homological nature of entropy". In: *Entropy* 17.5 (2015), pp. 3253–3318.